13. IX.

Experiment - 03.

- **AIM:** Convolutional Neural Network (Any one from the following) • Use any dataset of plant disease and design a plant disease detection system using CNN • Use MNIST Fashion Dataset and create a classifier to classify fashion clothing into categories.

- **THEORY:**

(1). **Introduction.** -

- Convolutional neural networks are a specialized class of deep learning models designed to process and analyse visual data such as images and videos. Unlike traditional neural networks, CNNs exploit spatial heirarchies in images by using convolutional operations, enabling them to automatically learn important visual features such as edges, textures, shapes and patterns.

- In agriculture, plant diseases pose a serious threat to crop yeild, food security, and farmer livelihoods. Early and accurate disease detection is essential to minimize crop loss and reduce excessive use of pesticides.

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- Manual inspection by experts is time consuming, costly and often impractical for large scale farms. Deep learning based classification provides an efficient and scalable solution to this problem.

- In this study, a convolutional Neural network is designed and implemented to automatically detect and classify plant diseases using leaf images from a publicly available plant disease dataset.

(2). Dataset Description -

- A commonly used dataset for plant disease detection is the Plant Village dataset, which contains labeled images of healthy and diseased plant leaves.

- Dataset Characteristics -

  - Datatype - RGB images.
  - Image Content - Plant leaf images.
  - Classes - Multiple disease classes and healthy classes.

  - Image size - Resized to a fixed dimension (eg 128 x 128, 224 x 224)

- Dataset Split -
  - Training set.
  - Validation set.
  - Test Set.

- The dataset contains images of crops such as tomato, potato, corn, apple, grape and others with different disease categories.

(3). Need for CNN in plant disease detection -

- Traditional image processing techniques require hand-crafted features and struggle with complex backgrounds and variations in lighting, texture and leaf shape.
- CNNs overcome these limitations by automatically learning discriminative features from raw images.

- Advantages of CNNs -

- Automatic feature extraction.
- Translation invariance.
- High accuracy in image classification.
- Robust to variations in scale and illumination.

(4) Architecture of Convolutional Neural Network -

A CNN consists of multiple layers arranged in a heirarchical structure to extract increasingly complex features from images.

4.1 Convolutional Layer.
- Applies learnable filters to input images.
- Extracts low-level features such as edges and corners.

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4.2. Activation function (ReLU). -

- Introduces non-linearity.
- ReLU (Rectified linear Unit) is commonly used

4.3. Pooling Layer. -

- Reduces spatial dimensions.
- Helps in reducing computational cost.
- Max pooling is widely used.

4.4. Fully connected layer -

- Combines extracted features.
- Performs classification based on learned representations.

4.5. Output Layer -

- Uses softmax activation for multiclass classification.
- Outputs probability for each disease class

(5). Image Processing -

Image processing is essential to ensure uniformity and improve model performance.

Preprocessing steps -

1. Image resizing to fixed dimensions.
2. Normalization of pixel values.
3. Data augmentation (rotation, flipping, zooming).

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4. Splitting into training and testing sets.

Data augmentation improves generalization and reduces overfitting.

(6). Model Training -

- Loss Function - Categorical cross entropy.
  - Suitable for multiclass classification.
- Optimizer. - Adam optimizer.
  - Provides faster convergence and stability.
- Evaluation Metrics - Accuracy
  - Loss.

During training, CNN learns disease-specific visual-patterns such as spots, discoloration, and texture changes on leaves.

(8). Model evaluation -

- The trained model is evaluated on unseen test images to measure its performance.
- CNN effectively identifies disease specific patterns
- Performance improves with deeper architectures.
- Data augmentation enhances robustness.

(9). Applications -

- Smart agriculture systems
- Mobile based plant disease diagnosis.

- Precision Farming.
- Crop monitoring and yeild optimization.
- Decision support systems for farmers.

(10) Result and Analysis.

- The CNN acheives high classification accuracy.
- Healthy leaves are clearly distinguishable from diseased leaves.
- Misclassifications mainly occur between visually similar diseases.
- Feature maps reveal progressive abstraction from edges to disease patterns.

(11). Pros and Cons.

Pros.	Cons.
1. High accuracy in disease detection.	1. Requires large labeled datasets.
2. Reduces dependency on agricultural experts.	2. Computationally intensive training.
3. Fast and scalable solution.	3. Performance may drop for unseen diseases.
4. Supports early disease diagnosis.	4. Sensitive to image quality and lighting conditions.

* CONCLUSIONS : Thus we successfully used CNN for plant disease detection.

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